

Medical Thoracoscopy With vs Without Prior Artificial Pneumothorax for Patients With Minimal or Absent Pleural Effusion



Kaige Wang, MD; Liang Zhou, MD; Min Zhu, MD; Wei Zhang, MD; Zhengguang He, MD; Xiaowu Tan, MD; Xing Luo, MSc; Lingfeng Min, MD; Feng Xu, MD; Jun Zeng, MSc; Hao Qin, MD; Jun Wang, MD; Huizhen Liu, MD; Dan Liu, MD; Panwen Tian, MD; Luca Richeldi, MD; Weimin Li, MD; and Fengming Luo, MD

BACKGROUND: Thoracoscopy guidelines recommend inducing artificial pneumothorax before medical thoracoscopy in patients with minimal or absent pleural effusion. Single-arm studies have demonstrated that nonartificial pneumothorax approaches reduce operative time and complication rates compared with artificial pneumothorax techniques in these patients. However, there is a lack of trials comparing the effectiveness and safety of performing artificial pneumothorax vs not performing it in these cases.

RESEARCH QUESTION: For patients with minimal or absent pleural effusion, is nonartificial pneumothorax (non-AP) noninferior to artificial pneumothorax (AP) in terms of the pleural access success rate?

STUDY DESIGN AND METHODS: In this multicenter randomized noninferiority trial, patients with minimal or absent pleural effusion requiring medical thoracoscopy were randomized 1:1 to either an AP group or a non-AP group. The primary outcome was the pleural access success rate, with a noninferiority margin of -10% (non-AP group minus AP group). Secondary outcomes included pathological confirmation rates, complication rates, operation length, air leak duration, drain removal time, chest pain scores, and 90-d mortality.

RESULTS: A total of 204 participants were equally allocated to the AP group ($n = 102$) and the non-AP group ($n = 102$). The pleural access success rate was 95.0% in the non-AP group and 78.4% in the AP group (difference, 16.6% [one-sided lower 97.5% CI, 7.6%]; $P < .001$ for noninferiority). Complications occurred in 14.9% of non-AP patients and 17.6% of AP patients (difference, 2.7% [95% CI, -12.9% to 7.3%]; $P = 0.589$).

INTERPRETATION: For patients with minimal or absent pleural effusion, artificial pneumothorax is not necessary before performing medical thoracoscopy.

CLINICAL TRIAL REGISTRATION: This study has been prospectively registered in the Chinese Clinical Trial Register, No. ChiCTR2000038708. CHEST 2026; 169(1):269-279

KEY WORDS: artificial pneumothorax; medical thoracoscopy; pleural diseases

FOR EDITORIAL COMMENT, SEE PAGE 18

ABBREVIATIONS: AP = artificial pneumothorax; IQR = interquartile range; MT = medical thoracoscopy; non-AP = nonartificial pneumothorax; PAS = pleural adherence scoring; VAS = visual analog scale

AFFILIATIONS: From the Department of Pulmonary and Critical Care Medicine (K. W., L. Z., M. Z., D. L., P. T., W. L., and F. L.), West China Hospital, Sichuan University; the State Key Laboratory of

Respiratory Health and Multimorbidity (K. W., L. Z., M. Z., D. L., P. T., W. L., and F. L.), West China Hospital, Sichuan University; the Laboratory of Pulmonary Immunology and Inflammation (L. Z., M. Z., and F. L.), Department of Respiratory and Critical Care Medicine, Frontiers Science Center for Disease-Related Molecular Network,

Take-Home Points

Study Question: For patients with minimal or absent pleural effusion, is nonartificial pneumothorax (non-AP) noninferior to artificial pneumothorax (AP) in terms of the pleural access success rate?

Result: In this study, non-AP achieved a significantly higher pleural access success rate than AP.

Interpretation: Our results suggest that for patients with minimal or absent pleural effusion, AP is not necessary before performing medical thoracoscopy.

Medical thoracoscopy (MT) is the reference standard diagnostic approach for investigating unknown exudative pleural effusion or pleural abnormalities by obtaining pleural effusion or tissue samples,¹⁻⁵ and it requires adequate pleural space to place a trocar and perform thoracoscopic pleural inspection.⁶ Cases with minimal or absence of pleural effusion, often

accompanied by pleural adhesions, increase the difficulty of MT because of the limited pleural space.⁷⁻⁹ Thoracoscopy guidelines recommend inducing artificial pneumothorax (AP) before MT in such cases to expose the pleural cavity and to locate approaches.^{10,11} Single-arm studies have indicated that in patients with pleural effusion, nonartificial pneumothorax (non-AP) reduced the risks of complications and shortened the surgical duration while maintaining comparable pleural access success rates.¹²⁻¹⁴ However, no randomized clinical trials have compared the efficacy and safety of non-AP with AP in patients with little or no pleural effusion. Therefore, this prospective study aimed to confirm whether non-AP is noninferior to AP in terms of pleural access success in patients with minimal or absent pleural effusion. Pathological confirmation rates, complications, length of operation, chest air leak duration, drain removal duration, chest pain visual analog scale (VAS) score, and 90-day mortality were compared as secondary outcomes.

Study Design and Methods

This prospective randomized multicenter noninferiority trial was performed at 6 hospitals in China between July 2021 and November 2023. The study protocol was approved by the West China Hospital of Sichuan University ethics committee [approval no. 2020(881)] and

by the local ethics committee of all the participating centers. All the participants provided written informed consent. The study has been prospectively registered in the Chinese Clinical Trial Register (Identifier: ChiCTR2000038708), and the protocol has been published. Results were reported according to CONSORT 2010 guidelines.¹⁵

Participants

Eligible patients were those aged at least 18 years with undiagnosed pleural disease, with minimal (maximum, 3-cm depth when seated) or no pleural effusion, and who were scheduled for medical thoracoscopy. Patients were ineligible if they had any of the following conditions before randomization: (1) complete absence of pleural space due to adhesions¹⁶; (2) inability to tolerate medical thoracoscopy due to (a) lateral decubitus intolerance; (b) cardiovascular and hemodynamic instability; (c) significant (associated with symptoms or heart failure) pulmonary hypertension; (d) aortic dissection; (e) myocardial infarction; (f) refractory cough; (g) ongoing respiratory failure; (h) bleeding tendency (ie, platelets < 60 × 10⁹/L, or international normalized ratio > 1.2); (i) short expected survival or poor systemic condition; or (j) cutaneous infection, metastasis, or rib fracture around the port insertion site; (3) consent withdrawal; and (4) enrollment in other research studies.

To assess the presence and severity of pleural adhesions, we used the pleural adherence scoring (PAS) system

West China Hospital, Sichuan University, Chengdu; the Department of Pulmonary and Critical Care Medicine (W. Z., H. Q., and J. W.), First Affiliated Hospital, Second Military Medical University, Shanghai; the Center of Critical Care Medicine (W. Z.), First Affiliated Hospital, Second Military Medical University, Shanghai, China; the Department of Pulmonary and Critical Care Medicine (Z. H. and F. X.), Suining Third People's Hospital, Suining; the Department of Pulmonary and Critical Care Medicine (X. T. and X. L.), Second Affiliated Hospital, Hengyang Medical School, University of South China, Hengyang; the Department of Pulmonary and Critical Care Medicine (L. M.), Subei People's Hospital of Yangzhou University, Yangzhou; the Department of Pulmonary and Critical Care Medicine (J. Z.), Clinical Medical College and First Affiliated Hospital of Chengdu Medical College; the Center of Biostatistics, Design, Measurement and Evaluation (CBDME), Department of Clinical Research Management (H. L.), West China Hospital, Sichuan University, Chengdu, China; the Division of Pulmonary Medicine (L. R.), Fondazione Policlinico Universitario A. Gemelli IRCCS, Rome; and the Division of Pulmonary Medicine (L. R.), Università Cattolica del Sacro Cuore, Milan, Italy.

K. W., L. Z., M. Z., and W. Z. contributed equally to this manuscript.

CORRESPONDENCE TO: Fengming Luo, MD; email: fengmingluo@outlook.com

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from the Sonographic and Biological Pleurodesis Indicators of Malignant Pleural Effusion (SIMPLE) trial.¹⁷ We evaluated 9 zones across the fifth, sixth, seventh, or eighth intercostal spaces along the anterior axillary, mid-axillary, and posterior axillary lines. Using dynamic thoracic ultrasonography examination, we scored pleural sliding as present (1 point), questionable (2 points), or absent (3 points). We interpreted scores as follows: PAS = 9-20, nonextensive adhesions; PAS = 21-25, extensive adhesions; and PAS = 26-27, complete adhesions (1 of our exclusion criteria). The classification criteria were based on the SIMPLE trial standards and practical considerations (a score of 25 represents two zones with questionable pleural sliding or one zone with pleural sliding). Patients in the lateral decubitus position with anechoic areas (indicating **loculated** pleural effusion) were not subjected to this scoring system.

Randomization and Masking

Patients who met the eligibility criteria were randomly assigned to undergo MT, either with an AP (AP group) or without (non-AP group), in a 1:1 ratio. Randomization was performed by block randomization, stratified according to participating center. A single randomization list was created for all participants. Allocation concealment was maintained using sequentially numbered, sealed, opaque envelopes with a randomization number. Owing to the research design, only the patients could be masked to the treatment allocation.

Patient Follow-Up

All patients were followed up at 90 d after the procedure by the research team via telephone interviews or outpatient visits. The follow-up assessments included surgical wound status, postoperative complications, and mortality.

Procedures Protocol

Preoperative thoracic ultrasound assessment and localization of approach: Before the MT procedure, each patient underwent a thoracic ultrasound examination in the healthy lateral decubitus position, using an Esaote SpA ultrasound machine (Siemens Acuson Free-style) equipped with convex (3.5 MHz) and linear (7.5 MHz) transducers (e-Fig 1). The method for selecting pleural access points prioritizes the anechoic zone, followed by areas with pleural sliding sign, and then zones with questionable pleural sliding. If multiple areas exhibit these characteristics, preference is given to those in the seventh intercostal space.

Establishment and evaluation of AP: In the AP group, an AP was induced before MT. The entire procedure was conducted by a team of thoracoscopists at the respiratory endoscopy center. In patients with an anechoic area sign at the pleural access point, pleural effusion was extracted via thoracic puncture, followed by a guidewire and single-lumen venous catheter placement. A concentric puncture cannula was used in patients without an anechoic area at the pleural access point. The needle was inserted vertically into the intercostal muscle layer, and the core was removed. A pistonless syringe containing 5 mL of sterilized saline was connected to a blunt needle. The needle was advanced perpendicular to the chest wall. A rapid decline in saline levels indicates entry into the pleural cavity. On successful puncture, a central venous catheter was inserted 4 to 10 cm into the chest. The guidewire was then removed and the catheter was fixed. Using the catheter, filtered air was slowly and gently injected into the pleural cavity until resistance was encountered or a maximum volume of 300 mL was reached.⁷ Radiography or thoracic ultrasonography was used to confirm pneumothorax induction. Success was defined as the presence of a pneumothorax band on radiography or specific ultrasound signs, including absent pleural sliding, lung points, or the M-mode stratosphere sign (e-Fig 2).

The procedural steps of medical thoracoscopy: Throughout the procedure, all patients remained in a state of conscious sedation and analgesia with propofol and remifentanyl, as monitored by continuous ECG. For patients in the non-AP group, the step of establishing artificial pneumothorax was not performed. Instead, for these patients, once the pleural access point was determined, a 1-cm incision was carefully made after local anesthesia with 2% lidocaine. A trocar was inserted into the pleural cavity following simple blunt dissection, and the adhesion band around the incision site could subsequently be dissected bluntly. For patients in both groups, the parietal, diaphragmatic, and visceral pleura were examined with a semirigid thoracoscope (Olympus LTF-240) with video assistance. Biopsies were performed with biopsy forceps through a semirigid thoracoscope. At least 8 pleural biopsy specimens were collected from each patient.

Treatment of patients with failed procedures: Procedure success was defined as gaining access to the pleural space, conducting a thoracoscopic pleural inspection, aspirating fluid (if visible), or obtaining a biopsy. If

the procedure failed, CT- or ultrasound-guided pleural biopsy was the first option for obtaining pleural specimens. For patients with failure to establish the AP and who decline CT- or ultrasound-guided pleural biopsies, performing MT via a non-AP approach can be an alternative option.

Study Outcomes

Medical thoracoscopy primarily enables diagnosis of unknown exudative pleural effusion or pleural abnormalities through obtaining biological specimens (pleural fluid or tissue samples). Therefore, the primary outcome was defined as the pleural access success rate, calculated as the proportion of successful entries into the pleural cavity that allowed the collection of pleural effusion or tissue specimens divided by the total number of patients in the group.

Secondary outcomes included pathological confirmation rate, complications, length of surgery, chest air leak duration, time of drain removal, chest pain VAS score, and mortality within 90 d postoperatively. Pathological confirmation rate was defined as the proportion of pathological diagnoses confirmed by pleural effusion or specimens acquired through MT divided by the total number of patients in the group. Complications were categorized as AP or MT complications. Complications secondary to AP included pleural reactions, thoracic bleeding, lung lacerations, mediastinal or subcutaneous emphysema, and air embolisms. Complications secondary to MT included pleural reactions, thoracic bleeding, lung lacerations, subcutaneous emphysema, postoperative fever, poor wound healing, and tumor metastasis through a chest wall incision. Thoracic bleeding was defined as a persistent drainage output exceeding 200 mL/h for 2 to 4 consecutive hours.¹⁸ Length of operation was the time spent on invasive procedures during the operation. Air leak duration was defined as the time from the end of the operation until the bubbles ceased in the chest drainage bottle. Drain removal time was defined as the time from the end of the operation until the drainage tube was removed. The chest pain

VAS assessment was completed within 48 h after surgery.

Statistical Analysis

The sample size for the trial was calculated on the basis of the primary outcome of “pleural access success rate.” Previous research has reported an 87% success rate for AP,⁶ whereas a retrospective analysis at our hospital revealed a 90% success rate for the non-AP-locating approach. Following consultation with clinical experts, we established a noninferiority margin of –10% for successful pleural access (non-AP group minus AP group). We calculated that a sample size of 204 patients, with 102 in each group, would provide the trial with 80% power to determine whether the non-AP localization approach was noninferior to the AP approach with respect to the success of pleural access. The calculation was performed with PASS 15 software (NCSS Statistical Software), accounting for a 10% postoperative dropout rate and the necessity for random allocation.

For comparative analysis between groups, χ^2 tests for categorical variables and independent *t* tests or Mann-Whitney *U* tests for continuous variables were employed, depending on their distribution. The primary analysis used a CI approach, wherein noninferiority was established if the lower bound of the two-sided 95% CI for the between-group difference exceeded the negative value of the predetermined noninferiority margin. Two-sided 95% CIs have a lower boundary equivalent to a 1-sided 97.5% CI. We reported between-group difference estimates and 95% CI for all secondary outcomes. Failed cases were assessed using descriptive statistics to provide insights into the characteristics and potential reasons for unsuccessful procedures. Subgroup analysis based on the thoracic ultrasound findings were used to further identify the efficacy of non-AP and AP treatments in patients with variable degrees of adhesions. To address missing data in secondary analysis, we adopted a complete case analysis approach. All analyses were performed with IBM SPSS Statistics software, version 29.

Results

Patients

From July 2021 to November 2023, 1,130 patients were assessed for eligibility, of whom 926 were excluded (e-Table 1) and 204 were eligible and randomly assigned to either the non-AP group (*n* = 102) or the AP group (*n* = 102; e-Table 2). Of the 204 enrolled patients,

1 refused to continue participating in the study (Fig 1). Primary outcome data were available for all patients (100%). During the 90-day follow-up period, 1.5% of patients (*n* = 3) were lost to follow-up for secondary outcomes.

The demographic, clinical, and surgical characteristics of the patients at baseline were similar between the 2 groups

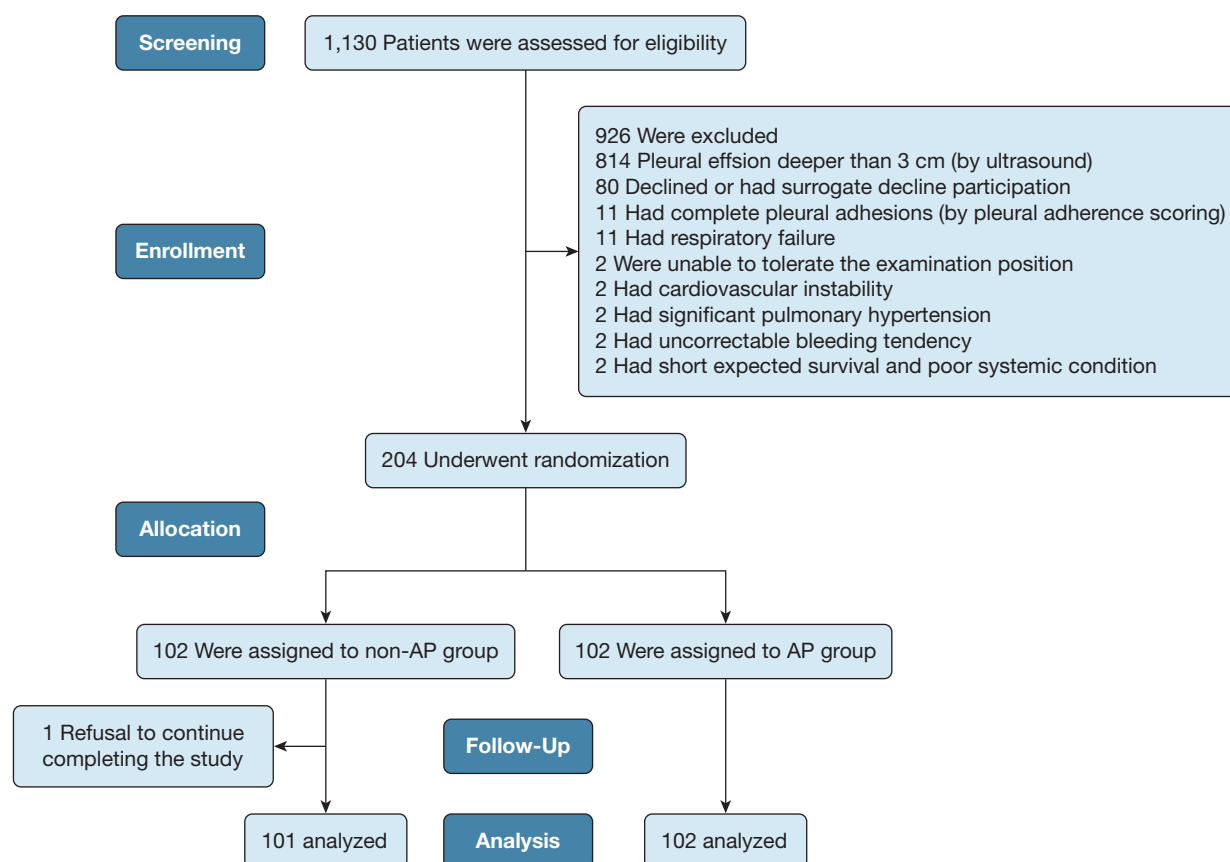


Figure 1 – Patient enrollment, randomization, follow-up, and analysis. AP = artificial pneumothorax; non-AP = nonartificial pneumothorax.

(Table 1). General characteristics and clinical features of the participants are shown in e-Table 3. The patients were predominantly male (127 patients, 62.6%) with a mean age of 58.2 ± 12.5 years. The mean BMI was 23.2 ± 3.3 , and 79 patients (38.9%) had a smoking history.

At baseline, most patients ($n = 117$; 57.6%) presented with pleural sliding at the pleural access point, and 61 patients (30.0%) showed signs of an anechoic area at the pleural access point. The pleural effusion depth was 15.0 mm (interquartile range [IQR], 10.0-20.0). The primary diagnosis was undiagnosed pleural effusion (168 patients; 82.8%). In addition, undiagnosed pleural nodules were identified in 23 patients (11.3%), and pleural thickening was identified in 12 patients (5.9%).

Primary Outcome

Pleural access success was achieved in 96 of 101 patients (95.0%) in the non-AP group and in 80 of 102 patients (78.4%) in the AP group (e-Table 4). The between-group difference was 16.6% (1-sided lower 97.5% CI, 7.6%;) (Table 2, Fig 2), indicating noninferiority ($P < .001$).

Secondary Outcomes

The pathological confirmation rates were similar between the non-AP and AP groups (76.0% vs 75.0%, respectively; Table 2; detailed results in e-Table 5), with a between-group difference of 1 percentage point (95% CI, -11.7 to 13.8; $P = .873$). The proportion of malignancies was 58.9% (43 of 73 patients) in the non-AP group and 61.7% (37 of 60 patients) in the AP group, with a between-group difference of -2.8% (95% CI, -19.5 to 13.9%). Pleural fluid analysis results are presented in e-Table 6. The complication rates were similar between the 2 groups, 14.9% ($n = 15$) in the non-AP group and 17.6% ($n = 18$) in the AP group (difference, 2.7 percentage points; 95% CI, -12.9 to 7.3; $P = 0.589$). The proportion of complications related to AP was 7.8% in the AP group, and no AP-related complications were observed in the non-AP group. The non-AP group had a shorter operation time of 32.4 min (IQR, 26.8-40.0) compared with 39.9 min (IQR, 34.8-47.2) in the AP group. The chest air leak and drain removal durations were also shorter in the non-AP group, with between-group differences of -1.5 h

TABLE 1] Baseline Characteristics of Patients

Characteristic	Non-AP (n = 101)	AP (n = 102)	Total (N = 203)
Age, y	58.1 (11.4)	58.4 (13.6)	58.2 (12.5)
Sex, male	63 (62.4)	64 (62.7)	127 (62.6)
BMI, kg/m ²	23.2 (3.1)	23.2 (3.5)	23.2 (3.3)
Smoking history	38 (37.6)	41 (40.2)	79 (38.9)
Comorbidity ^a			
Cardiovascular disease	32 (31.7)	36 (35.3)	68 (33.5)
Respiratory system	12 (11.9)	13 (12.7)	25 (12.3)
Endocrine system	18 (17.8)	14 (13.7)	32 (15.8)
Digestive system	10 (9.9)	8 (7.8)	18 (8.9)
Immune system	0 (0.0)	4 (3.9)	4 (2.0)
Symptom duration before randomization, d ^a	48.0 (21.0-97.0)	36.5 (20.0-90.0)	40.0 (20.0-97.0)
Symptoms ^a			
Fever	19 (18.8)	16 (15.7)	35 (17.2)
Fatigue	3 (3.0)	3 (2.9)	6 (3.0)
Edema of both lower limbs	2 (2.0)	2 (2.0)	4 (2.0)
Weight loss	7 (6.9)	11 (10.8)	18 (8.9)
Cough	51 (50.5)	50 (49.0)	101 (49.8)
Expectoration	11 (10.9)	9 (8.8)	20 (9.9)
Hemoptysis	1 (1.0)	0 (0.0)	1 (0.5)
Dyspnea	50 (49.5)	51 (50.0)	101 (49.8)
Chest tightness	30 (29.7)	33 (32.4)	63 (31.0)
Chest pain	35 (35.7)	38 (37.3)	73 (36.0)
Sore throat	1 (1.0)	0 (0.0)	1 (0.5)
Backache	3 (3.0)	5 (4.9)	8 (3.9)
Abdominal pain	3 (3.0)	1 (1.0)	4 (2.0)
Abdominal distension	5 (5.0)	3 (2.9)	8 (3.9)
Proposed location of approaches			
Right side	68 (67.3)	65 (63.7)	133 (65.5)
Pleural adherence scoring of affected lung ^b	9.0 (9.0-16.0)	9.0 (9.0-17.0)	9.0 (9.0-16.0)
Ultrasound signs at the pleural access point ^c			
Pleural sliding sign	58 (57.4)	59 (57.8)	117 (57.6)
Questionable pleural sliding sign	12 (11.9)	13 (12.7)	25 (12.3)
Anechoic area	31 (30.7)	30 (29.4)	61 (30.0)
Depth of pleural effusion, mm	15.0 (10.0-20.0)	16.6 (10.0-20.0)	15 (10.0-20.0)
Primary diagnosis			
Undiagnosed pleural effusion	78 (77.2)	90 (88.2)	168 (82.8)
Undiagnosed pleural nodules	15 (14.9)	8 (7.8)	23 (11.3)
Undiagnosed pleural thickening	8 (7.9)	4 (3.9)	12 (5.9)

Data represent No. (%), median (IQR), or mean (SD) unless otherwise stated. Percentages may not total 100 because of rounding. Summary statistics are reported for nonmissing values. AP = artificial pneumothorax; IQR = interquartile range; non-AP = nonartificial pneumothorax.

^aComorbidity, symptom duration, or symptoms were reported by the patient, and 1 patient could present with several comorbidities or symptoms.

^bPatients in the lateral decubitus position with anechoic areas (indicating loculated pleural effusion) were not subjected to the pleural adherence scoring system.

^cFor each patient, only 1 ultrasound sign was recorded, with priority given to the presence of an anechoic area (indicating pleural effusion), followed by an assessment of the pleural sliding sign.

TABLE 2] Primary and Secondary Outcomes

Outcome	Non-AP	AP	Absolute Difference (95% CI)	P Value
Primary Outcome^a				
Pleural access success rate	96/101 (95.0)	80/102 (78.4)	16.6 (7.6 to 25.7)	< .001
Secondary Outcomes				
Pathological confirmation rates ^b (excluding failed cases)	73/96 (76.0)	60/80 (75.0)	1.0 (−11.7 to 13.8)	.873
Malignancy	43/73 (58.9)	37/60 (61.7)	−2.8 (−19.5 to 13.9)	.746
Benign disease	30/73 (41.1)	23/60 (38.3)	2.8 (−13.9 to 19.5)	.746
Complications ^c	15/101 (14.9)	18/102 (17.6)	−2.7 (−12.9 to 7.3)	.589
Complications related to AP				
Pleural reaction	NA	1/102 (1.0)	NA	
Thoracic bleeding	NA	0/102 (0)	NA	
Lung laceration	NA	3/102 (2.9)	NA	
Postoperative subcutaneous emphysema	NA	3/102 (2.9)	NA	
Air embolism	NA	1/ 102 (1.0)	NA	
Complications related to MT	15/96 (15.6)	10/80 (12.5)	3.1 (−7.1 to 13.4)	.554
Pleural reaction	3/96 (3.1)	2/80 (2.5)	0.6 (−4.3 to 5.5)	1.000
Thoracic bleeding	1/96 (1.0)	0/80 (0.0)	1 (−1.0 to 3.1)	1.000
Lung laceration	4/96 (4.2)	3/80 (3.8)	0.4 (−5.4 to 6.2)	1.000
Postoperative subcutaneous emphysema	3/96 (3.1)	5/80 (6.3)	−3.2 (−9.5 to 3.2)	.471
Postoperative fever	1/96 (1.0)	0/80 (0.0)	1.0 (−1.0 to 3.1)	1.000
Poor wound healing	2/96 (2.1)	0/80 (0.0)	2.1 (−0.8 to 4.9)	.501
Tumor metastasis through chest wall incision	1/96 (1.0)	0/80 (0.0)	1.0 (−1.0 to 3.1)	1.000
Length of operation, min	32.4 (26.8 to 40.0)	39.9 (34.8 to 47.2)	−7.5 (−10.5 to −4.4)	< .001
Chest air leak duration, h	26.1 (6.8 to 44.0)	27.6 (14.8 to 48.0)	−1.5 (−7.7 to 4.6)	.428
Time of drain removal, h	51.4 (14.0 to 72.0)	56.5 (24.0 to 72.0)	−5.0 (−19.0 to 9.0)	.151
Chest visual analog scale score ^d				.663
Mild	61/101 (52.5)	57/102 (41.2)	NA	
Moderate	15/101 (14.9)	13/102 (12.7)	NA	
Severe	3/101 (3.0)	0/102 (0.0)	NA	
Death in 90 d	1/99 (1.0)	2/101 (2.0)	−1.0 (−4.3 to 2.4)	.573

Data are presented as n/N (%) or median (IQR). AP = artificial pneumothorax; IQR = interquartile range; MT = medical thoracoscopy; NA = not applicable; non-AP = nonartificial pneumothorax.

^a $P < .001$ for noninferiority.

^bPathological diagnoses were established by MT pleural specimens.

^cComplications related to MT, length of operation, chest air leak duration, and time of drain removal were analyzed in patients of pleural access success. One patient could present with several complications. There were no complications related to AP in the non-AP group.

^dThe classification of the chest visual analog scale pain scores was defined as follows: 1-3, mild pain; 4-6, moderate pain; and 7-10, severe pain.

(95% CI, −7.7 to 4.6) and −5.0 h (95% CI, −19.0 to 9.0), respectively. Chest pain levels, measured using the VAS within 2 day after the procedure, showed no significant differences between the groups. Mortality within 90 day was 1% (1 of 99 patients) in the non-AP group due to tumor progression and 2% (2 of 101 patients) in the AP group, with 1 death due to myocardial infarction and the other due to tumor progression.

Subgroup Analysis

In the subgroup analysis based on PAS, the pleural access success rates varied between the non-AP and AP groups (Table 3). In the PAS ≤ 20 subgroup, the pleural access success rates were significantly higher in the non-AP group than in the AP group (100% vs 81.4%; $P < .05$). However, the severity of pleural adhesions observed during medical

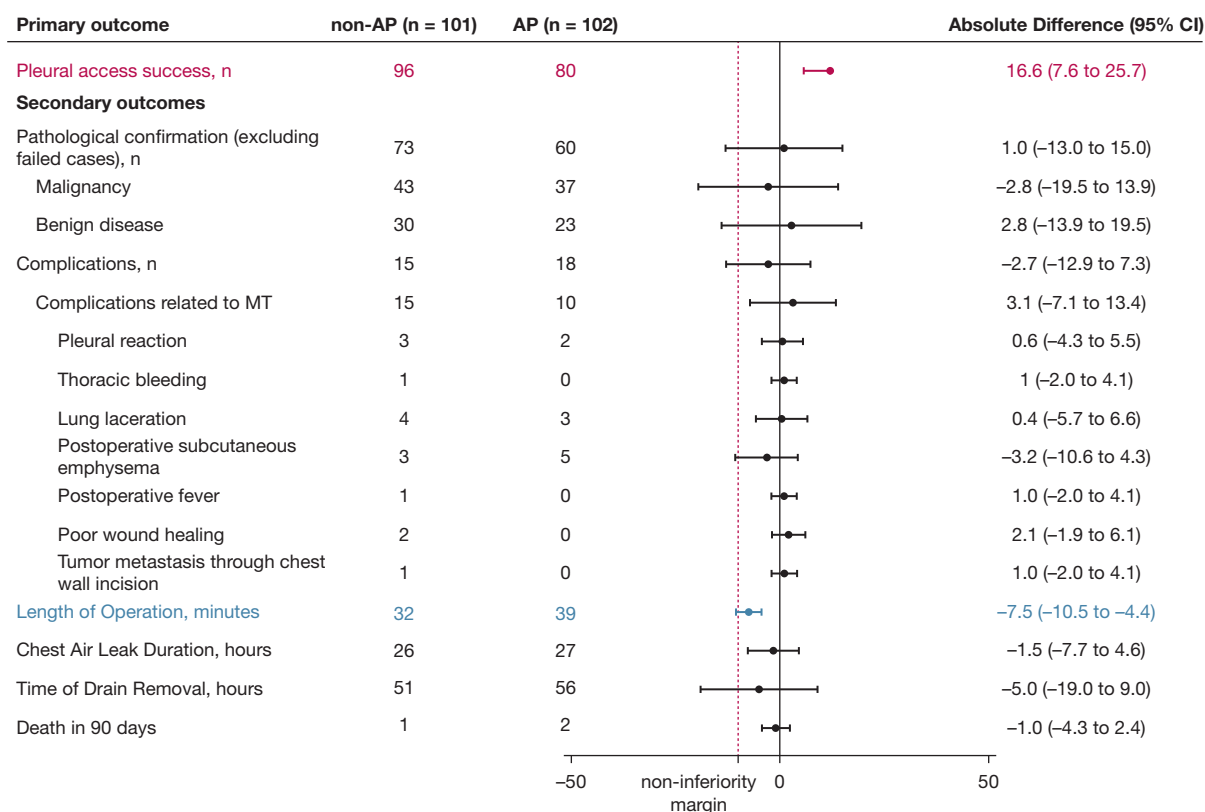


Figure 2 – Primary and secondary outcomes in the non-AP and AP groups. Absolute difference is presented as percentages, with 95% CI represented by a horizontal bar. Red signifies a statistically significant difference in the primary endpoint between the non-AP and AP groups, whereas blue denotes a statistically significant difference in the secondary endpoint between the two groups. AP = inducing artificial pneumothorax before medical thoracoscopy; non-AP = not inducing artificial pneumothorax before medical thoracoscopy.

thoracoscopy was not significantly different between the non-AP and AP groups.

Failed Patients

Failure in the non-AP group was due to severe adhesions (5 of 5 patients, 100%; e-Table 7), whereas, in the AP group, it was nearly all due to AP-related causes (21 of 22 patients; 95%). The procedure was terminated in 1 patient (5%) in the AP group due to a severe pleural reaction during medical thoracoscopy.

Patients who failed pleural access were biopsied according to the preset protocol. In the non-AP group, 2 patients (40%; e-Table 7) underwent CT-guided biopsy, and 3 patients (60%) underwent ultrasound-guided biopsy. In the AP group, 9 patients (41%) required non-AP methods, whereas 3 patients (14%) and 9 patients (41%) required CT- and ultrasound-guided biopsies, respectively. One patient (5%) in the AP group declined further invasive procedures and thus remained undiagnosed.

Discussion

In this randomized trial, we found that non-AP was noninferior to AP in achieving pleural access success rates among patients with minimal or absent pleural effusion. In the $PAS \leq 20$ subgroup, pleural access success rates were significantly higher in the non-AP group than in the AP group. However, given the small sample size and the few complications in each group, this conclusion should be interpreted with caution.

The results of this study revealed that in the AP group, the predominant cause of pleural access establishment failure was an unsuccessful AP procedure (accounting for 95% of cases). This was primarily attributed to the presence of pleural adhesions, which are common in patients with minimal or no pleural effusion. The AP establishment process involves slowly injecting filtered gas to separate adhesions. However, this method could address only a small portion of the adhesions, proving insufficient to adequately expose the pleural space required for medical thoracoscopy. Conversely, in the non-AP group, once the pleural access point was confirmed via preoperative ultrasound, a 1-cm incision

TABLE 3] Subgroup Analysis of Primary Outcome and Thoracoscopic Manifestations Based on Pleural Adherence Scoring

Characteristic ^a	Non-AP (n = 101)			AP (n = 102)		
	PAS > 20 (n = 12)	PAS ≤ 20 (n = 58)	Anechoic Area (n = 31)	PAS > 20 (n = 13)	PAS ≤ 20 (n = 59)	Anechoic Area (n = 30)
PAS	21.83 (0.72)	10.60 (2.88)	NA	22.15 (0.80)	10.54 (2.90)	NA
Pleural access success rate	8/12 (66.7)	58/58 (100.0) ^b	30/31 (96.8)	5/13 (38.5)	48/59 (81.4) ^b	27/30 (90.0)
Thoracoscopic manifestation						
No pleural adhesions	0/8 (0.0)	23/58 (39.7)	8/30 (26.7)	0/5 (0.0)	23/48 (47.9)	8/27 (29.6)
Localized pleural adhesions	0/8 (0.0)	25/58 (43.1)	9/30 (30.0)	0/5 (0.0)	19/48 (39.6)	9/27 (33.3)
Extensive pleural adhesions	8/8 (100.0)	10/58 (17.2)	13/30 (43.3)	5/5 (100.0)	6/48 (12.5)	10/27 (37.0)

Data are presented as n/N (%) or mean (SD). AP = artificial pneumothorax; NA = not applicable; non-AP = nonartificial pneumothorax; PAS = pleural adherence scoring.

^aNo pleural adhesions defined as no adhesions covering of the visual field under thoracoscopy. Localized pleural adhesions defined as adhesions covering 0% to 50% of the visual field under thoracoscopy. Extensive pleural adhesions defined as adhesions covering 50% to 100% of the visual field under thoracoscopy.

^b*P* = .002.

was made, allowing the operator to bluntly dissect local adhesions. This, in turn, achieved adequate exposure of the pleural cavity, which enhanced the pleural access success rate.

Moreover, the non-AP method allows the pleural cavity to communicate with the external environment, creating a pressure difference between the negative pressure in the pleural space and the atmospheric pressure. This results in the automatic formation of a pneumothorax in patients, thereby reducing the need for AP (e-Fig 3). Consequently, non-AP reduces operation time and diminishes the risk of severe complications associated with AP.^{6,14}

In addition, subgroup analysis revealed that the non-AP approach demonstrated higher or trending higher pleural access success rates compared with the AP method across various patient subgroups, including those with anechoic areas, PAS ≤ 20, and PAS > 20. In the PAS ≤ 20 subgroup, the non-AP approach enabled effective blunt dissection of adhesive tissue, adequately exposing the pleural space and significantly improving success rates (58 of 58 [100%] vs 48 of 59 [81.4%]). In the PAS > 20 subgroup, despite severe pleural adhesions limiting success for both approaches, the non-AP method achieved numerically higher success rates through blunt dissection techniques (8 of 12 [66.7%] vs 5 of 13 [38.5%]), but the small sample size limited statistical significance. Similarly, in the anechoic area subgroup, although some patients with loculated effusions failed artificial pneumothorax establishment, the non-AP approach facilitated pleural access through blunt separation of fibrous encapsulations, yielding improved success rates (30 of 31 [96.8%] vs 27 of 30 [90.0%]). However, statistical detection was also limited by sample size. This consistent performance across diverse subgroups indicates that the advantages of the non-AP approach are not limited by specific pleural characteristics, further supporting our primary research conclusion. Furthermore, PAS effectively identified the severity of adhesions; all patients with PAS > 20 presented with extensive pleural adhesions under medical thoracoscopy (13 of 13; 100%). Among patients with PAS ≤ 20, 90.6% (90 of 106) showed nonextensive adhesions under thoracoscopy. This finding further confirms the reliability of ultrasound in identifying adhesions based on the semiquantitative scoring system.¹⁹

There were slight differences between the results of our randomized study and those of the previous studies.

Previous cohort studies have reported relatively consistent rates of pleural access success ranging from 83.3% to 100% for non-AP, and from 87% to 89.6% for AP.^{6,12,14,20,21} These discrepancies may be attributed to several factors. First, most previous studies were retrospective cohort studies, which might have introduced selection bias and confounding factors that could have influenced the outcomes. In contrast, our study employed a prospective, randomized design that minimized potential biases and ensured a more balanced distribution of patient characteristics between the 2 groups at baseline. Second, the heterogeneity in patient populations and disease severity across single-arm studies may have contributed to variations in pleural access success rates. Our randomized trial minimized the impact of potential confounders by implementing rigorous inclusion and exclusion criteria, which resulted in well-balanced baseline characteristics between the 2 groups. This well-designed randomization process allowed for a more reliable and unbiased comparison of the effectiveness of AP and non-AP approaches.

This study has some limitations. First, the relatively small sample size of the subgroup analysis, especially for categories with a small number of patients (eg, those with PAS > 20), may undermine the statistical power needed to detect significant differences or potentially introduce bias. Nevertheless, the present study is 1 of the largest cohort studies to focus on this less common pleural effusion subtype. Second, the interpretation of thoracic ultrasound lacks unified guidelines. In this study, some measures were taken in advance to reduce error: all operators were well trained in thoracic ultrasound interpretation, uncertain cases were further assessed through discussions by research coordinators from multiple centers, and we employed a semiquantitative scoring system, PAS, with reference to the SIMPLE trial, to provide a more objective and standardized assessment of pleural adhesions. Third, owing to the nature of the surgery, this study adopted a single-anonymized design, which may introduce bias

and subjectivity in the evaluation. To minimize the impact of these confounders, metrics for objective evaluation were selected, standard operating procedures were established, and standardized training was conducted to ensure uniformity in the execution of procedures by different surgeons. In addition, we used a multicenter collaborative approach to increase the generality and credibility of the results.

Interpretation

Our study demonstrated that, among patients with minimal or absent pleural effusion, non-AP is noninferior to AP in terms of pleural access success rate, while maintaining comparable complication rates. These findings support the concept that non-AP is an effective and safe method that can streamline the MT procedure by eliminating the need for AP. Therefore, for patients with minimal or absent pleural effusion, AP is not necessary before performing MT.

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Dissemination to participants and related patient and public communities: The results of this work will be disseminated to the public through the press, social media, and conferences. The lay summaries of the study results will be communicated to study participants and relevant patient groups.

Additional information: The e-Figures and e-Tables are available online under “Supplementary Data.”

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